

ActInf Textbook Group ~ Cohort 5 ~ Session 6 (Chapter 3, Part 1) ~ 10/24/2

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 6 (Chapter 3, Part 1) ~ 10/24/2023 | 55:15

all right welcome everyone it's October
24th 23 and we're in our first
discussion of chapter
three for cohort 5 so before we go to
any specific question does anyone just
want to give any
first thoughts or Reflections on
three just where it is in the textbook
or um what's stuck with them or
what they remember from the
journey Oli want to give any first
thoughts or anyone
else yeah sure uh well chapter 3 as
opposed to chapter
2's um kind of bottomup approach or the
low road uh takes the more daunting or
uh let's put it this way the more
adventurous route of uh the high road uh
which is the one that's been
increasingly uh developed in uh the
recent years uh so it begins with the
notion of what it is it or let me put it
this way if we suppose
that quote unquote things exist in an
environment or in the World At Large
what
characteristics uh should they possess
to consider them as ontological things
in other words uh what character
characteristics should they possess if
we want to uh kind of re-evaluate or
remeasure them uh at a specific time
scale uh and uh consider them as cons
consistently the same thing so the idea
is if we put a kind of statistical

boundary between the internal states of that object or the thing and its uh external States or its environment then it would be possible to uh to underwrite the Dynamics of um of the of the internal states of the system uh at face of the changing environment given that Mark of blanket or this or the statistical boundary so in other words the internal states of the system um does not change directly um in a response to the changing environment but the relationship between those two states are kind of they kind of vicariously uh influence uh the external States vicariously influence the uh internal state so they don't have direct access but only through the blanket States or Mark of blanket so that's basically the idea for the a self-organized object um and uh it's it's uh one of the main main properties for any living thing or uh rather an idea for uh how life Works um I mean by life here I mean in a very broad term and not just necessarily biological life but it also encompasses any kind of artificial life as well such as xenobots is developed by Michael Levin's lab or any other uh organoids or um similar uh entities that's been increasingly uh appearing in uh many various disciplines and research areas thank you that's a great summary Kristoff then anyone else who raise their hands um I just wanted to ask because

the the concept of um Mar Blanca is sort of a separation between the internal and and external environment it always makes me think is that is isn't it really kind of talking about uh temporal SE separation that um ultimately what what really says that we have states that we can identify that are not directly influenced by uh by the environment but but ultimately the system is it is connected so um it's it's it's this this separation like you know eventually breaks down right the signal the signal gets through the barrier and finally influences the internal States there's a lot of pieces there first to last piece yes the signals do get through they do influence so there's causal propagation that can reify the blanket so blankets can exist as interface they do exist as the interface they also only exist over a certain spatial or temporal time scale and then you in the beginning described like time in a maravian sense and that's like a more common usage of the maravian property is through time like the past only influences the future through the present so then the current moment is a markup blanket in time so you could make a markup blanket around any Baye graph you write the Baye graph out you pick the node as internal that's the system of Interest internal states there is a base gra um there is a mark on blanket so that could be um through time you could have one variable through time and talk about maravian

property through time it could be
spatially embedded and then it could
refer to spatial temporal enclosure but
the concept does
relatively little abstract philosophical
work because by the time you define the
basian graph it's basically trivial or
deflationary where the blankets
are and yet it has attracted some of the
most heated
philosophical
discussion yeah go ahead
rere
so I want to try to connect
the formal
definition of the Markov blanket as
we're using it in this context
with what we humans seem to have and
maybe probably other animals as well uh
which is some intuitive notion of what
it is uh to be a
thing uh because we look at the at our
environment
and maybe some of it corresponds to to
these partitions that can be defined as
Mark blankets uh but we maybe see an
animal we see some system and we
identify this collection of entities or
properties this corresponds
to one
thing um and of course we can look at it
more closely and start to see more
individual things inside of that that
Collective but well we do have some uh
notion of of thingness that's quite uh
very
intuitive and uh to follow up on that is
uh I want to understand what kinds of

systems uh we can reasonably
describe uh through the idea of a mark
of blanket to give an example uh I've
heard CL fiston talk
about uh setting a a drop of ink
inside um maybe like some water a cup of
water and the Dynamics of that
um ink
drop not being maybe not being
describable or being
describable uh with a mark of blanket uh
perhaps the fact that the ink disperses
and
doesn't uh maintain
it um its coherence is what he means is
what is trying to to point out like that
means uh this isn't showing that
property of thingness and of uh
remeasuring that we we we see when we
think about a thing um so yeah I want to
get a sense for what's
what uh kinds of systems whether they're
biological or
not uh are we is it reasonable to
describe using that this the idea of a
markable blanket
yeah does anyone who have a hand raised
just want to speak directly to it
otherwise just keep it up and then we'll
get to another topic but first just what
can you model appropriately with a
Markov
blanket if anyone just wants to go for
it my understanding is that any it's
basically anything right because if it
didn't have that separation between self
and environment there wouldn't be
a distinction between environment and

the thing itself one uh the drop the example of the drop of ink is one that I actually really like because it shows that a drop of ink will dissipate inevitably but if fit with a living drop of ink just as a single cell yes it would be an constant state of dissipation but also the cell is able to extract uh quote unquote energy from the surroundings right by breaking down some molecules in the surroundings and using that energy to hold itself together and also to use the byproducts to make more components of itself so it kind of counters this dispersion over time um imagining that process to me is you know I'm a biologist and so on but it's still very very hard one example I really like is this bottle I have here oh my background is a little blurred but if I press this button here in this in this shaker I make a little Vortex right and uh it's kind of kind of cool because like it's making something out of nothing it's like you have just environment and you have an input of energy that set that makes a a low lower end tropy system right that demarcate itself from its environment which is made of the same thing as as this thing and all you need as a battery is dissipating a gradient right if you don't have the gradient you have no fluid Vortex and one thing on the that know they that that I don't know if somebody has an answer is uh what is a mark of does this thing have have a markco

Blanca just the kind of flame is and if
it does well it it has to right because
it's a thing separated from other
things what is the can we classify
different kinds of Micco blanks you know
like this is this is very transient it's
very thin it's so but that membrane the
membrane of a cell is a more much more
complicated blanket
um many many points just to very briefly
note them because it's again it's always
more time in between our discussions
like yes it is about the cell and the
ink here's the one of the earliest 2006
examples there's a snowflake so
snowflakes are getting whipped around
like your Vortex external energy is
going through the so what can you say
about a snowflake well it's frozen but
what could you say if there were some
snowflakes that also could preclude
melting so that's why it has to start
with a thing and a lot falls out from
you starting with a persistent thing or
a remeasured thing does a thing have a
markup blanket this is the Eternal
realist instrumentalist question the
realist language is yes it has a marov
blanket cells do active inference this
organism has a Precision estimate the
instrumentalist position is you can
model it with a marov blanket it's as if
it has a Precision variable and so
whether we say that it's about what the
territory is doing about what it has or
is versus what we could do classifying
different kinds yes absolutely transient
watering blankets these are the kinds of

things that many people are researching about because when you actually do have a system of Interest these kinds of questions are just immediately the things that you need to

ask um Jesse

uh or or yeah go ahead David

sorry okay

Jess may I just interject for a moment to add another angle sure this yeah okay so um another perspective for viewing a markup blanket um uh I mean whether or not it is ubiquitous or or just uh it only applies to specific things is that uh okay so theoretically it is possible to draw uh draw some statistical boundary between anything that is sufficiently separated from its environment but uh for the things that follow or that have a kind of linear causal structure uh to their Dynamics or to their interactions with the with the environment this markup blanket doesn't say anything specifically interesting about the

Dynamics so uh the significance of markup blanket uh is becomes more apparent uh when we we're dealing with the systems with nonlinear causal structures to their interactions with the environment so the more uh nonlinear their Dynamics the more uh interesting things markup blanket can say about uh the uh the the interaction of that system

and it can

uh somehow result in

non-trivial um

non-trivial elements of um describing of
the description of the system so uh as
Dalton St Vel has shown in one of his
paper week uh blankets uh it's not uh I
mean the question of markup blanket it's
not a question of whether or not but
it's a question of degree so we can have
uh a Continuum of uh
blanketed uh uh if you will of different
uh different degrees of strength for
that markup blanket so again for linear
systems we have very um trivial kind of
markup blankets and as the system get
get more complicated or more complex and
uh with higher dimensional um structure
a markup blank gets become increasingly
non-trivial thank you

oi um Jesse and then and then

Susan yeah that's really interesting

what you're saying Ali because I mean
the way that I've been viewing the
Markoff blanket is I know

that there is I guess a temptation to
extrapolate like a broader meaning about
existence and thingness but I'm just
kind of looking at it as a statistical
tool to you know delimit inputs and
right it's like it's a way of organizing
inputs and defining external States and
internal States because in the things as
they are the the number of variables of
inputs is really difficult to you know
quantify in that that broad causal chain
and so that's maybe where Ali that what
you're saying about the nonlinearity is
is really interesting um but in model
making where we are like investigating
specific input output relationships the

Markoff blanket seems to be very useful for us to be able to you know create that type of basian graph and organize that information great yeah and also like blanket only exists as part of the particular partition in relation to the rest that are being described so it's not like a blanket drawing exercise it's like a system modeling exercise and then taking different system of interest um as your internal State you just by definition get a marov blanket so hence it does little philosophical work by itself and yet has attracted some of the most attention so so um so aren't blankets um at isn't there blanket versus blanket in other words there's different scales and when you say partition that you know that kind of maybe are you when you say partition are you talking about um Markov blankets at different levels and and to you to kind of tie on to what Jesse was saying art Markoff blankets context specific okay a lot there let's think about a few more tangible pieces first here's the definition so we're talking about the relationships of variables not about the relationships of things in the world so any discussion of what a markup blanket is slash does as the technical concept it's not going to be supported or rejected by any feature of the world it's about the variables being related and then the partition is just the definition of what the markco blanket is

doing which is a set of nodes one or more that are intervening between two other sets that comes into play when modeling how variables are connected so if we made a model of uh temperature in the room and there was just one thermometer carrying forward through time we might have one model where every time Point can influence every other time point but then for long time series that's going to get out of control so a common modeling assumption is like well make it a Markov chain a one Markov chain where only the last time step influences the next time step so that was what was mentioned earlier like that's a a pure Markov blanket in time is that a claim that there isn't like time effects in the real world well not specifically but then you could at least use that and say actually we're finding that the one markov chain model Capt pretty much all the empirical phenomena so still that's not saying that there aren't some time effects in some setting but just saying that it's totally sufficient for that case so so one more question that for me ties it up is that when you say Markoff Chain versus the model I'm kind of relating a markof chain to a map yeah I think you can even go further the whole thing is a map we're only talking about cognitive cartography only map and then the marov chain is this Motif where you have a set of chained States hidden States through time with a

markov transition Matrix

But that's the one Markov property is that
is that horizontal

chain and so like you could have a
probability Matrix that is like on the
diagonal it has

things staying the same Sunny to Sunny
0.9 and then cloudy to cloudy 0.9 and
then the off diagonals is like the
transition between them so those are
some simple statistical models of
dynamic systems and a Markov chain is
when they're just chained forward in
time the Markov

blanket could be considered from a
static perspective again that's not to
say like organisms are or aren't Dynamic
just about the markup blanket you could

have a causal Network that is
static uh it's how Bert in his work in
55.0 today introduced the Bas graph

concept was with a static causal
chain just kind of contextual factors
interacting like a structural equation

model but in a Bayesian setting and then

you can imagine composing a Bas graph

that has um a spatial and a temporal

component or um a temporal and a

perceptual component or a temporal and a

perceptual and an active

component and then when we have a base

graph in

time with something like

perception and action then that is where

we get the partially observable

mdp but like Markov decision process

was a term before and outside of this

kind of Markov boundary as organism

boundary discussion marov decision
process is just referring to the marov
chain and making decisions on a Markoff
chain yeah it's an interesting top and
it's the heart of the high
road as as Ali mentioned because whether
we're from the outside talking about
remeasuring
something that's like so then there's
things that are static compared to how
we're remeasuring them there's things
that are changing faster than we can
remeasure them so then they're not being
detected as the same thing like gas
molecules in room but then there's some
spot where there is like a temporal
thickness in modeling
and then that could be seen from either
the
remeasurement the
experimental position which comes back
in chapter
nine looking like
this experimenter observing
something
or to persist in time that can be framed
as a remeasurement of homeostatic
attractors
rather than for
example the maximization of a reward or
utility
function it's like well now the zeroth
level task is to
remeasure in the homeostatic
range not to be rewarded by values in
the homeostatic range and then maximize
reward but to directly Target the
statistical measurement

any other topics or or quote to a
section that somebody wants to like go
to yeah yeah just this whole
relationship between um value and and
belief you know the relationship between
the value function and and um know the
the function of Prior beliefs and
surprise minimization I thought
was a really
interesting uh jump
that I'd love to talk more about or to
have some people's thoughts
on which pag section or anyone can just
give it a
thought
section

3.5 page page

53 yeah very related to the previous
discussion replaces the notion of value
with the notion of Prior
belief so we can contrast the sort of
imperative to be value and so so the the
the um the base assertion to be value
exists and then the imperative to be
that it has to be optimized or
maximized alternatively a sort of
foundational assertion can be the thing
exists which might be kind of a neat
assertion because we might only be
interested in those systems that do
fulfill that and then the imperative
would be the kind of more boring and
first

level be that thing whatever it is be as
it is as um

and instead of setting up even
anticipatory so so that that seems to to
more closely apply to like okay the rock

is just sitting there the bird is just
flying okay things just kind of do their
own thing but then that's the
variational free energy so then the next
question is well then how do you
actually
get snowflakes that fly and resist
melting cells that don't just dissipate
like an ink drop how do you actually get
something to both be what it is and have
a preference or realize its preference
or have agency in realizing its
preferences and that's where the
expected free energy comes
in again making no assertions about the
mechanism that any given system actually
does to to fly and avoid danger but
giving a statistical framework for the
bird
watcher so that they can say something
statistical about the behavior that
they're
seeing and then the other side of the
coin is like the mechanistic and and the
and that's the low
road H how is the bird of the bat
avoiding danger well it's not actually
doing matrix
multiplication but it's a whole question
to actually investigate how and that's
the low road for for each system but the
high road is kind of the starting point
for discussion because without the high
road it's like an admission that it's a
purely hypothetical
space and then even that exists in a
sense
inks and then anyone

else I just wanted to give an example
about this um in my work often people
act against their goal
because the I was going to say value but
the the intrinsic reward
for proving their unconscious prior
belief

I am the one who suffers I am the one
that everyone hates somehow the
organisms values that higher this prior
belief of identity I am that thing than
actually achieving a goal that they set
for themselves so to to me intuitively
that makes a lot of sense that swap
there because I imagine that has to do
with the

um deep affect that comes with prision
evaluating your prior is good like my
old systems work my old priors are valid
so there is a high there
around good affect I may be suffering I
may be losing but I was right all along
can't take that away from

me yeah lot lot to say there so in the
affective inference work they linked the
psychological phenomena of affect to
Precision estimation so if things were
more variable than expected that was
negative AFF effect if things were more
precise it was positive a effect and if
things were as precise as expected as
variable as expected it was neutral
balance and so if there's an empirical
track record of something being a
certain way like I've gotten out and
gotten \$1 each day I'm the kind of
person who gets \$1 and then now there's
an affordance that is empirically not

been taken then there's these different mechanisms to explore like Precision on policy or Precision on sensory inputs maybe the three is like surprising and so then it's like less cognitive overh good or just more natural flow to go back into an empirical previous way of doing it and but then multiple models are constructed it's not just like well a toy model was made and now this is like you know people are going to be using this in a clinic it's like multiple models are iterated in in a lot larger and a slower setting over the years and then more and more is understood about computational Psychiatry um Kristoff then anyone else um I find it uh interesting to to think about this uh you know um this effectively functional of functional beliefs as as a sort of you know this is the This Is The Meta function that we're trying to optimize and and and I often find it that when I talk to people about for instance the uh the notion of Free Will um I like to say that we don't really have free will because you know if you consider really your circumstances very carefully you will find out after um you know uh sufficient analysis that there's usually the correct uh way to act and uh you're only it only feels like you have this free will because you're acting under uncertainty and you're making a decision for for for the first time so it it feels like you're doing something because there is there is this place

there is a wiggle room but
um you know this this this really nicely
fits with this you know notion of Prior
beliefs that sort of uh those prior
beliefs make you what you what you are
you are a thing because you have prior
beliefs and I like to use the uh the
term a Aesthetics um because that really
determines you know um uh kind of the
kind of values you're going to pursu is
the things that the things that you care
about the uh the the the things you find
beautiful the things the way the way you
want to be that's kind of Aesthetics but
the the the process um of you know so
you can you can in some in some way you
can choose that but the process of how
you're going going to get to your
preferred States that's
predetermined by the by the by the
physics of reality right so that's uh
and and that's where the kind of like
the this disension between having having
free will having the the wiggled room
having Aesthetics that you can choose
having the prer beliefs that you can
that that you kind of have to well you
can't you can choose them um if you
learn how then you you get to adjust
your Precision parameters right it's
just a m matter of practice to get to
get the root access to to the mind to be
able to do that um and then and then
everything happens is kind of
automatically from that point on because
if you're a free energy minimizing agent
you always know what to do next find the
the the the juiciest pocket of negative

entropy Goble it up and and uh go about your

business yeah I mean saying that a thing is or or even has priors is the definition of realism certainly a stance that people have asserted but also people can use the same formalism on that exact same system of Interest or a different system of interest in support of an instrumentalist

claim so the framework can be used to advance a realist claim but at least from my perspective that doesn't make mathematics an assertion about any specific system or how any specific thing is it's literally to the exclusion of that that we have a general framework

Susan um so I I actually put about four questions in the in the notes I didn't know if you were going to try to get to the questions um in uh shall you look at it from here or should I go to the chapter or core five questions here uh the questions yeah yeah Y and and I guess specifically yeah what what's which one what word to what word to look up to find it

affordances

okay um and and that yeah kind of actually pulls on both the thank you the question in terms of inks and Kristoff's um discussion in terms of you know are these can these be learned

um okay great thank you for adding them so Ali explains in the intro video at the top of the page Markov blanket is a boundary in the state space not

necessarily a spatial temporal
boundary came up today as well what
allows the agent to be statistically
separated from its boundary as
observed through its
blanket so I take it that the blanket is
a lens through which the agent observes
environment yeah lot lot of ways that
that that could be
taken but yeah if we if we kind of
associate why with observation and the
chain of perception yes the
inbound
dependencies across the blanket do map
to like a lens and then also it's a
great um connection as some will
probably see that actually in the
category Theory like the lens there's
probably even a deeper meaning of this
too but for so we won't go into it but
short answer
yes would this lens be defined by their
so the outgoing
dependencies from the blanket so
literally the other side of perception
action outbound dependencies the space
of the possible actions is defined by
affordances but if the hidden state were
a person then what their affordances
would be you know the observations of
the other person so these are not tags
on the world it's not like this object
is an observation this object is an
action it's like in this map with this
as my internal State then this
dependency can be read as perception or
as
action affordances according to Gibson

must be learned yeah I mean there's really two things here there's kind of like Gibson Aly like what did this specific person think about this word which I'm always all for but the short answer is affordance is being used in a slightly different way in active inference not to refer to the direct agentic perception of an action capacity but a little bit more from a modeler's perspective as just simply the action capacities whether or not they're perceived without making any statement on whether they have to be learned or not we can model a pendulum with active inference so obviously the affordance concept doesn't need learning you can just model it without learning does the distinction between active learning and active inference have to do with these affordances so yeah this was a distinction that that one paper in the Discord mentioned short answer here is you could have two agents with the same set of affordances Move up down left right and then you could set up the parameters differently so one of them sought to confirm its expectations about sensory another one sought to reduce its uncertainty maximally about world States so those are all the parametric fine-tunings but it doesn't require a difference in affordances but it might require difference in how the affordances are being taken that's how you would know that the behaviors were different at

all you could also make a model where there were different affordances so that's for situational modeling and then the question becomes very obvious are internal State parameters referring to the edge of the marov blanket State space not exactly the internal State parameters are everything about the the system of interest that isn't the interface so it's like the shell of the egg is the Marco blanket and then the inside is the internal States because like when you take a photo you're just going to see the shell so that's just the outer interface um with perception action across the blanket and then internal State parameters are everything about that thing that isn't the shell and then together the thing is internal plus blanket because that's what moves around relative to the does anyone else have any like kind of other other takes on any of these questions yeah Oli please uh yeah so uh there's another uh rather simplified um representation of markup blanket with regards to uh the probabilistic dependencies of the different states uh in uh the physics of sentience U presentation by K fris uh so if you search for uh this topic in YouTube there are several different presentations but around uh the the time four minutes or something you can see a clear uh representation of markup

blanket which shows how exactly external States internal States sensory States and active States relate to each other according to their probabilistic uh dependencies so I think that can be quite helpful

cool also yeah a lot of other lot of other pieces these are all these are all great great

questions and it's one reason why there's such an emphasis on making the generative models because there is such a difference between the abstract relationships purely within the actim fonty just the sort of conceptual relationships and then when mapping it to a system of Interest it's kind of like a 90 degree rotation with what you're connecting up connecting ACM terms to specific things and then when this is all accounted for that's all accounted for then that's the generative model but the textbook is describing like the pure relationships with a um stream of examples and boxes and kind of just different um pieces different

discussions another interesting thing is like the high roads describing the imperative but the low road is there's might be multiple ways to realize it so just again to separate out how it's being done from why it's being done um

Verena thank you go for

it hi I have a question and it might be um uh connecting to Chris of the uh question of um free will there is a

sentence in chapter three and I don't know where I'm so sorry but um what it says is that there is this um um when you maximize actually the uh free energy where you keep your options open until you have and I guess does it mean until I have enough evidence that I uh can State the belief and would it be a demonstration of Free Will that's my question that's a great great question it's kind of cool it comes up twice in the low road so kind of as a previously identified pattern of basian Statistics just having nothing to do with um modeling cybernetics and then here the maximum ENT y principle of Janes the imperative to explain things as accurately as possible but also keep options open Oli do you want to give a thought on this uh so yeah I think uh the question of Free Will related to active inference you see because um as far as I understand it um active inference is a kind of framework uh that puts agency at its Center Stage right uh so uh the minimal definition of agency for for us to uh endow or or confer agency to any system uh can be uh put it put in its most minimal sense in this way so if uh if there is more than one way to respond to a stimulus uh and and uh the other condition would be uh if the the system or the agent has a way to choose among uh those responses then uh we can

probably say uh it has some minimal sense of agency so in this sense I believe active inference models or um or the agents that's modeled with um markup blankets in active in in active inference Frameworks uh they have indeed this kind of minimal agency to them but if we go a little bit deeper than that and U consider uh the deeper generative models and deeper hierarchies then uh we would come across agents with beliefs about their beliefs H and we can of course continue that ad infan theoretically so beliefs about beliefs about beliefs and so on uh and uh the more deeper the generative model gets uh the more sentient or let's say uh probably more conscious the agent will become but in any case even for the most simple case of uh active particles we can um describe agency with regards to these minimal definitions so I believe uh active inference and basian mechanics put uh agency or rather Free Will um as some of its core tenants tenants for uh describing those systems so it is written to the core of active inference uh this conception of agencies and uh it's something that we cannot take for granted uh even if that kind of agency is not uh soal real agency but just a kind of apparent agency or in other words a kind of agency that appears as if the system have has so um yeah I I'm not sure how uh uh how IT addresses the question of whether there is uh whether or not there

is kind of free will for these kinds of Agents but um I think uh it can be uh seen as some kind of U minimal framework to um address these questions as well great I'll offer another position which is the maximum entropy principle also applies to gas molecules in the room and the temperature so I think even in this discussion we've heard some um positive and negative attributions of of free will an otive inference but then the question that that we can still circle around is well what's the relationship between the technical and analytical maximum entropy principle connected to all this other stuff for example the constrained maximum entropy principle being dual to the F more recently being found out but how does any of those principles relate to how some body mobilizes an account of a certain biological system for example that's even a different kind of question Susan yeah well along those lines I was just reading something about metacognition and um and some studies done um that uh that relate um the difference it made when um when students were actually taught how to think about their beliefs differently in the um and so I think that that for me that kind of brings in the the belief updating and the learning um you know as part of agency because even though I have different options I may you know my I'm assuming that you know the Markoff blanket can also retract depending on my

state um and you know that's going of um
hollow out opportunities
um I can't see my affordances even
though they're there would that
be would that be um you a am I pulling
those threads together correctly I
guess yeah that's great thank you just
so we can hear from everybody in the
last minutes um Brena then Kristoff then
Jesse um and I have another uh question
it's page

58 um the first sentence furthermore
active inference is largely coherent
with idomoo

Theory uh which states that action
starts with an imaginative process uh
has to been some work done what this uh
imaginative process is or um or what
what's part of the process and how it is
um yeah is there any any research group
do you know anyone or does anyone know
about um this kind of imaginative
process which triggers the
action that's a great question I think
you'd find that in essentially all
prospective ACM generative models you
could interpret it coherently with idoor
like if you chose to associate
um policy selection with ideas and then
the policy that gets
selected would would have Arisen from a
imaginative compositional process that
triggers
action there may also be people working
more on the um psychological side but
not I mean you can look it up or or post
in Discord or continue to explore but
but I think that it's a cool question

topic

because uh um I mean imaginative process

is a a process viewed from a human not

really from maybe a cell or or something

I guess I don't know that's uh that's

why I was like a bit triggered by by

that uh expression so um I wasn't sure

about what it was addressed here but

thank you for for the clarity I

will yeah it's like we we we come to it

from so many different way

well what if a computer program makes

new compositions is that

imaginative not that doesn't need to be

the claim that um a a human process is

the same as a computer process but if

they can be modeled with the same

formalism then from an instrumentalist

perspective it's just like saying well

we could measure a squirrel and a bird

with the same meter

stick that that makes limited claims

about different animals and how similar

and different they

are Jesse and then anyone

else yeah I I like the really thinking

about that relationship between

variational free energy and maximum

entropy uh sort of in the risk

management framework in a way it's it's

it's it's like engaging in this you know

it isn't just a matter of like

prediction error minimization but also

risk managing

predictions um in a certain

respect so that that that topic just

made me think like

that yeah it would be a risk to over or

underfit if you underfit the distribution you'd be leaving information out you wouldn't have as much awareness as you could if you overfit it would be fallacious in a different way so you'd want to be at the right distribution given what you knew yeah it's an interesting framework

totally agreed um Resa then Kristoff um hi uh so uh going back to um something Jesse raised about section 3.5 on the active inference side where it begins by describing uh how Behavior results in reinforcement learning versus uh active inference I was just thinking from a practical perspective has there do you know of any examples where um the simple reinforcement uh learning examples have also been implemented in code uh to demonstrate how the active inference approach to that would be like the card po card pole example or some some something um yeah Cod tangible cool the code page you'll find several dozen um applications and especially the work of um Alex shance Norse AED Connor Hines others uh Baron millage those tend to directly tackle and juxtapose non-active inference machine learning and and we've had many live streams on the mountain car the multi-arm Bandit several other kind of classical settings that make the comparison pretty um clear Kristoff and then anyone else with a last

word

uh yeah I just want to say about this
you know keeping keeping your your
options open it always reminds me of
this um fantastic uh thing that the
inestimable Terence McKenna said in one
of his Raves that you know he he was
asking this question like why us why is
the mushroom revealing the secrets of
the universe to us and he said like well
uh because I I don't believe in anything
I I I entertain ideas and I I try to
apply it definitely in my life is that
you know whenever I think I'm convinced
that I I understand something that that
that this this is true and that is not
true um I look back to that and also to
the teaching of of the Buddha and and
think to myself like no the answers are
never really that simple that's like the
question of free will simply might might
not have a binary answer it's just simp
as simple as that it's it's it's
complicated it's a process and we may
not be able to determine one way or
another because it simply might not be
um uh determinable that way the fact
that the syntax permits us to make a
statement doesn't mean it's a valid
statement to make um so that's uh
keeping your options open yeah that's a
that's that's an important thing to
meditate on um you know every day what
do I believe
nothing I gather evidence for um for the
existence of
ideas lot of different perspectives to
take great

times well thank you everybody we will
come back next week in the earlier time
for oh will be

Halloween chapter 3 a spooky

Adventure okay um costumes are

[Laughter]

optional thank you everybody see you

next time thank you everyone bye bye bye

thanks everyone bye